

NORTHGATE 5 ESE, ND, USA

WMO: 727540

Lat: 48.9676N

Lon: 102.1702W

Elev: 1842

StdP: 13.74

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 94084

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.6	-18.7	-28.5	1.2	-22.7	-23.3	1.6	-17.9	24.7	14.5	22.7	12.8	6.4	N/A	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	87.0	66.9	83.9	66.4	80.6	64.9	71.3	81.5	69.2	79.1	67.1	77.2	8.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	110.1	77.1	65.7	101.5	75.2	63.5	93.9	72.4	36.2	81.9	34.5	78.8	32.6	77.1	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.9	16.7	-32.5	93.9	4.7	4.2	-35.9	97.0	-38.7	99.5	-41.3	101.8	-44.8	104.9		
(5)			-31.3	75.7	3.5	2.8	-33.8	77.7	-35.9	79.3	-37.8	80.9	-40.3	82.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	9.7	7.7	23.6	38.1	51.7	61.2	66.3	64.6	55.5	41.1	25.5	11.6		
	DBStd	23.61	14.62	14.28	14.41	10.69	8.03	5.75	5.28	6.08	8.04	8.81	12.39	13.85		
	HDD50	5963	1250	1184	819	371	76	3	0	0	35	299	735	1191		
	HDD65	9971	1715	1604	1282	807	416	137	47	83	299	741	1184	1656		
	CDD50	1665	0	0	2	14	129	340	504	452	200	23	1	0		
	CDD65	199	0	0	0	0	5	23	87	70	14	0	0	0		
	CDH74	2902	0	0	0	14	152	349	996	1013	363	15	0	0		
	CDH80	808	0	0	0	1	30	59	267	343	106	2	0	0		
(14) Wind	WSAvg	7.4	9.0	8.0	8.3	8.2	7.9	6.9	5.8	5.6	6.4	7.2	7.5	7.6		
(15) Precipitation	PrecAvg	17.3	0.6	0.4	0.6	1.1	2.5	3.5	2.9	2.0	1.4	1.1	0.6	0.6		
	PrecMax	23.2	1.3	2.3	1.3	2.4	5.9	6.3	8.1	3.5	3.5	2.9	2.3	1.6		
	PrecMin	12.7	0.2	0.0	0.1	0.2	0.4	1.3	0.7	0.7	0.4	0.0	0.0	0.0		
	PrecStd	2.9	0.4	0.5	0.4	0.6	1.3	1.3	1.5	0.8	0.8	0.7	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	40.5	64.8	74.7	84.5	85.8	90.5	92.5	87.9	75.3	60.9	43.5		
		MCWB	36.2	36.1	50.3	54.5	60.4	67.3	71.9	66.7	65.1	57.9	49.0	36.4		
	2%	DB	37.4	36.5	55.7	67.5	77.9	81.1	86.0	87.6	83.0	67.8	52.5	37.8		
		MCWB	33.8	33.4	45.2	50.2	57.8	64.8	70.1	66.3	62.3	53.8	43.9	33.6		
	5%	DB	34.7	32.4	48.0	62.6	74.0	78.0	82.8	83.7	78.4	62.0	46.9	34.3		
		MCWB	32.0	30.4	41.8	47.2	55.6	62.9	69.1	66.1	60.7	49.9	40.4	31.3		
	10%	DB	31.0	28.7	41.3	57.0	69.2	75.0	79.7	79.8	71.8	56.2	41.7	30.2		
		MCWB	29.0	27.1	36.7	44.5	53.4	61.5	66.5	64.6	57.9	47.5	37.2	28.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	37.2	51.2	55.2	64.4	72.1	76.7	71.9	69.4	59.6	49.4	37.9		
		MCDB	41.4	42.2	60.3	72.1	76.1	79.0	83.4	83.4	81.6	73.9	60.3	43.9		
	2%	WB	34.3	34.3	45.8	50.4	60.9	67.7	72.6	70.0	64.4	55.8	44.2	34.5		
		MCDB	37.3	37.3	54.1	64.2	73.9	77.1	81.7	81.6	78.2	64.3	51.1	38.5		
	5%	WB	32.5	31.8	42.2	47.5	57.8	65.1	70.2	68.0	61.6	51.5	40.1	32.2		
		MCDB	35.1	33.8	49.4	59.9	69.4	74.6	79.6	79.9	72.5	61.0	45.0	35.4		
	10%	WB	29.8	28.4	37.5	44.7	55.5	63.0	68.2	65.9	59.2	48.3	36.9	29.4		
		MCDB	32.1	30.0	42.0	54.4	65.2	72.0	77.2	77.5	68.4	56.1	40.3	31.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	18.1	19.0	23.1	24.3	21.9	23.7	26.3	25.0	21.4	18.6	16.6		
		MCDBR	19.1	17.7	28.1	34.0	32.3	27.9	27.8	32.3	35.5	32.2	26.7	20.3		
		MCWBR	16.7	14.1	18.2	18.5	14.5	14.1	14.3	14.4	17.2	18.1	17.1	15.3		
	5% WB	MCDBR	20.0	15.9	27.8	31.6	25.9	23.4	25.0	27.7	28.4	30.3	24.3	18.6		
		MCWBR	17.4	13.1	18.5	18.0	12.9	13.3	14.5	13.5	14.9	17.9	16.5	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.255	0.280	0.328	0.342	0.359	0.368	0.379	0.338	0.292	0.257	0.236		
	taud		2.356	2.397	2.473	2.399	2.410	2.400	2.397	2.345	2.432	2.535	2.501	2.437		
	Ebn at Noon		261	291	300	292	291	285	281	272	274	273	262	258		
	Edh at Noon		21	25	28	34	35	36	36	36	30	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		478	827	1232	1567	1844	1964	2110	1707	1269	776	460	341		
	RadStd		45	80	104	103	129	137	99	120	108	84	63	33		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page